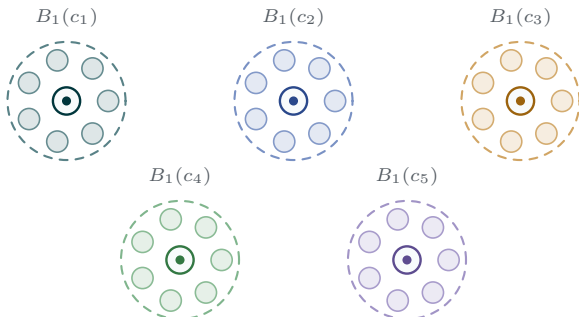


Perfect One-Error Correction

For $[7, 4, 3]$, each radius-one ball has $1 + 7 = 8$ words.



$$|\mathcal{C}| = 2^4 = 16$$

$$|B_1(c)| = 1 + 7 = 8$$

$$16 \cdot 8 = 2^7$$

The balls tile $\mathbb{F}_2^7$: every received word has a unique nearest codeword.